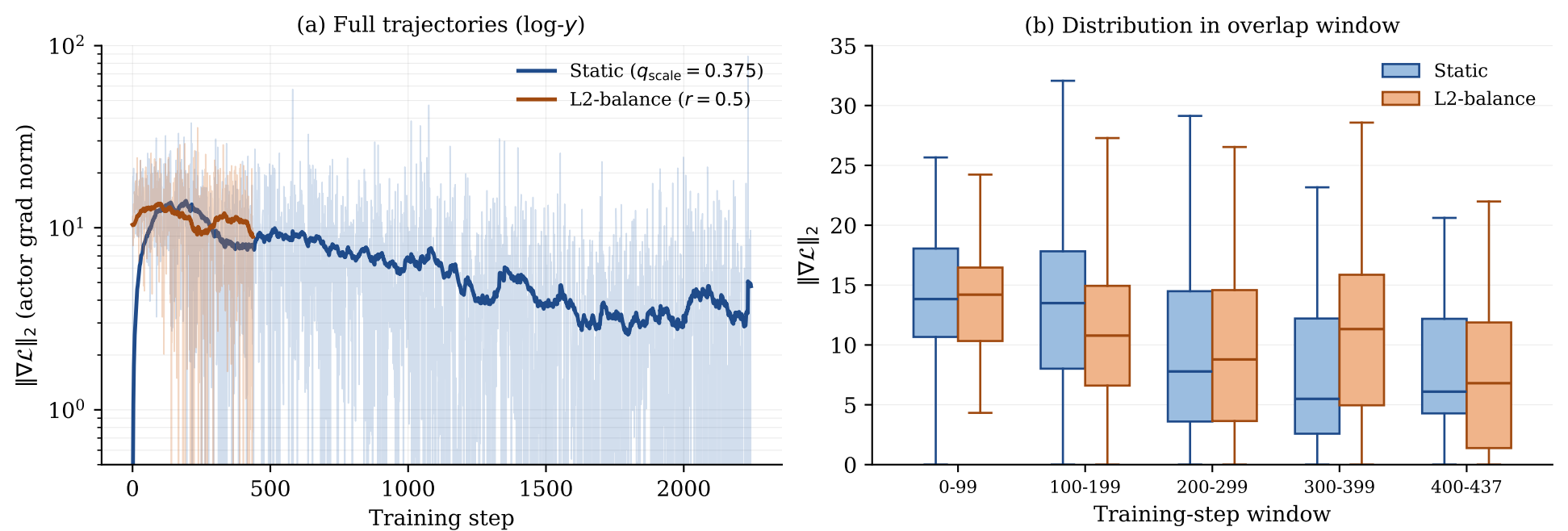

(a) Full trajectories (log-y)
Static ($q_{\text{scale}} = 0.375$)
L2-balance ($r = 0.5$)
$\|\nabla \mathcal{L}\|_2$ (actor grad norm)
10^2
10^1
10^0
0
500
1000
1500
2000
Training step
(b) Distribution in overlap window
Static
L2-balance
$\|\nabla \mathcal{L}\|_2$
35
30
25
20
15
10
5
0
0-99
100-199
200-299
300-399
400-437
Training-step window